

QUIZ ON ELECTRICITY

DATE: May 27, 2026

TOTAL MARKS: 35

STUDENT NAME: _____

ROLL NO: _____

SEC: _____

GENERAL INSTRUCTIONS:

[EXAM

STRUCTURE

REQUEST]:

Please arrange

questions into

distinct sections

as follows:

- Section A

(Multiple Choice

Questions):

Generate exactly

5 questions,

each worth 1

marks.

- Section B

(Short

Questions):

Generate exactly

2 questions,

each worth 5

marks.

- Section C

(Diagram/Graph-

Based

Questions):

Generate exactly

2 questions,

each worth 5

marks.

- Section D

(Numerical

Problems):

Generate exactly

2 questions,

each worth 5

marks.

The total exam

questions MUST

equal exactly 11

and overall

marks MUST

sum to exactly

35.

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[ADDITIONAL
USER
GUIDELINES]:Đ
Generate for 1.5
hour.
1. Read all
questions
carefully before
answering.
2. Write your
answers neatly
in the space
provided.
3. Verify all code
syntax or
reasoning steps
where requested.

SECTION A: MULTIPLE CHOICE QU ESTIONS

Attempt all
questions. Each
question carries
1 mark. Choose
the best option.

1. What is the SI unit of electric current?

[Easy] [1 Mark]

- (a) Volt
- (b) Ampere
- (c) Ohm
- (d) Watt

2. Which of the following is a good conductor of electricity?

[Easy] [1 Mark]

- (a) Rubber
- (b) Plastic
- (c) Copper
- (d) Wood

3. Ohm's Law states the relationship between which quantities?

[Moderate] [1 Mark]

- (a) Current, Power, Resistance
- (b) Voltage, Current, Resistance
- (c) Voltage, Power, Energy
- (d) Current, Time, Charge

4. What happens to the total resistance when resistors are connected in series? [Moderate] [1 Mark]
- It decreases
 - It increases
 - It remains the same
 - It becomes zero
5. A 60W light bulb is connected to a 240V supply. What is the current flowing through it? [Hard] [1 Mark]
- 0.25 A
 - 0.5 A
 - 1 A
 - 4 A

SECTION B: SHORT QUESTIONS

Attempt all questions. Each question carries 5 marks. Provide concise answers.

- Explain the difference between series and parallel connections of resistors, providing one advantage of each. [Moderate] [5 Marks]
- Describe the heating effect of electric current and give two practical applications. [Hard] [5 Marks]

SECTION C: DIAGRAM /GRAPH-BASED QUESTIONS

Attempt all questions. Each question carries 5 marks. Draw neat diagrams or interpret graphs as required.

- Draw a circuit diagram showing a battery, a switch, two resistors in series, and an ammeter measuring the total current. [Easy] [5 Marks]
- The V-I graph for a metallic wire at two different temperatures, T1 and T2, is shown. Both graphs are straight lines passing through the origin. The line for T1 has a steeper slope than the line for T2. Which temperature is higher and why? [Moderate] [5 Marks]

SECTION D: NUMERICAL PROBLEMS

Attempt all questions. Each question carries 5 marks.

*Show all
your
working
steps
clearly.*

1. Three resistors of $4\ \Omega$, $6\ \Omega$, and $12\ \Omega$ are connected in parallel. Calculate the equivalent resistance of the combination. **[Moderate]** [5 Marks]
2. An electric heater of resistance $8\ \Omega$ draws $15\ \text{A}$ from the service mains for 2 hours. **[Hard]** [5 Marks]
Calculate the rate at which heat is developed in the heater and the total energy consumed.